

UNIT I

Introduction to Artificial Intelligence

Artificial Intelligence: From Everyday Technology to Intelligent Systems

Understand & respond to information | Recognize patterns | Learn from experience |
Reason about choices | Solve problems & plan actions | Make goal-directed decisions

“How can a machine perform a task that normally requires some form of intelligence?”

What Will We Learn?

This unit takes us from the basic idea of Artificial Intelligence to the way an AI problem can be represented for computational solving.

- Introduction, history, and evolution of Artificial Intelligence
- Characteristics and types of Artificial Intelligence
- AI vs Machine Learning vs Deep Learning vs Data Science
- Intelligent systems and the AI lifecycle
- Applications of AI across major real-world domains
- AI problem formulation and state-space representation
- Problem characteristics, search space, solution space, performance measures
- Overview of AI-based problem-solving approaches

Learning path: Understand AI → Understand intelligent systems → Understand AI problems → Represent problems for solving

Consider a few systems you may already use

Machines That Appear Intelligent

System	Typical Intelligent Task
Navigation system	Recommends a suitable route
Voice assistant	Interprets a spoken request
Recommendation system	Predicts useful content
Medical AI system	Analyzes information for decision support
Face recognition	Identifies visual patterns
Autonomous system	Responds to changing conditions

These systems work in different domains. What connects them is their ability to use information to produce a meaningful response or action.

What Do We Mean by Intelligence?

Human intelligence involves several abilities:

- Learning from experience
- Understanding a situation
- Reasoning about available information
- Planning future actions
- Solving unfamiliar problems
- Making appropriate decisions

Artificial Intelligence studies how some of these capabilities can be represented and implemented in computational systems. The goal is not to reproduce every aspect of human intelligence — the focus is on building useful systems with intelligent capabilities.

What Makes a Task Intelligent?

A task becomes interesting from an AI perspective when the system must do more than perform one completely fixed operation. The challenge increases when:

- More than one action is possible
- The environment can change
- Information is incomplete or uncertain
- Patterns must be recognized
- The best response depends on context
- The system must work toward a goal

Example: A fixed program can always display the same route. An AI-based system may need to consider the current situation and choose a suitable route.

What Is Artificial Intelligence?

Definition: Artificial Intelligence (AI) is the field of computing concerned with developing systems capable of performing tasks that involve aspects of intelligent behavior.

Common AI capabilities include

- Learning
- Reasoning
- Planning
- Perception
- Problem solving
- Decision making

AI is therefore defined by the intelligent task being performed, not by the physical appearance of the machine.

Why Do We Need Artificial Intelligence?

Many real-world problems cannot be handled effectively using only fixed instructions. AI can be useful when systems need to:

- Process large or complex amounts of information
- Recognize patterns that are difficult to identify manually
- Respond to changing conditions
- Choose between multiple possible actions
- Support decisions using available evidence
- Solve problems where the exact response is not known in advance

Fixed rule

Condition → predefined response

AI-based system

***Information → interpretation → decision
→ response***

AI in Everyday Digital Systems

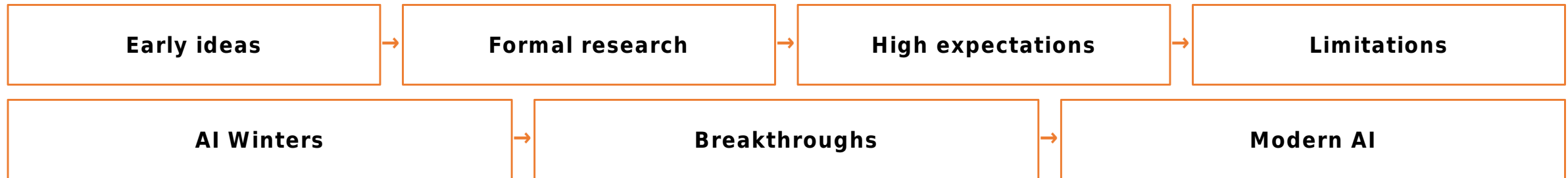
Artificial Intelligence is already integrated into many digital services.

Navigation	Route recommendation based on current conditions
Entertainment	Personalized content recommendations
Voice assistants	Language interpretation and response
Email	Detection of unwanted messages
Security	Recognition of unusual behavior or patterns
Online services	Prediction of preferences and next actions

AI often works behind the interface. Users may interact with the result without seeing the underlying intelligent process.

Why Study the History of AI?

Artificial Intelligence did not appear suddenly with modern applications. Its development has passed through several stages:



- How AI emerged as a research field
- Why early approaches faced difficulties
- How expectations changed
- Why major breakthroughs mattered
- How the field reached its present form

AI history shows that technological progress is shaped by both successes and limitations.

Early Foundations of Artificial Intelligence

“Can aspects of human intelligence be represented and performed by machines?”

Early research explored ideas related to

- Automated reasoning
- Problem solving
- Symbolic representation
- Game playing
- Language
- Knowledge-based systems

These early efforts established many of the questions that continue to influence AI research today. The field gradually moved from philosophical discussion toward computational investigation of intelligent behavior.

1950 — The Turing Test

In 1950, Alan Turing proposed the Turing Test as a way to think about machine intelligence.

“Can a machine produce responses that appear sufficiently intelligent during interaction?”

The idea focuses on the machine's observable behavior, rather than whether the machine physically resembles a person.

Why it matters: The Turing Test helped turn machine intelligence into a concrete topic for scientific discussion. It raised an enduring question: how should intelligent behavior from a machine be recognized?

1956 — Dartmouth Conference

The Dartmouth Conference of 1956 is an important milestone in the formal development of Artificial Intelligence as a research field. Researchers began exploring whether aspects of intelligence could be described and implemented computationally.

Early interests included

- Problem solving
- Reasoning
- Language
- Knowledge representation
- Game playing
- Machine-based decision making

The event helped establish Artificial Intelligence as a distinct area of research rather than only a collection of ideas from different fields.

Early AI — Ambition and Expectations

Early Artificial Intelligence research created considerable optimism about the future of intelligent machines. Researchers explored:

- Symbolic reasoning
- Automated problem solving
- Knowledge-based systems
- Game playing
- Automated reasoning

Expectations

Machines would soon perform many forms of intelligent behavior.

Practical capability

Existing systems could solve only limited and carefully defined problems.

That gap became an important part of AI's history.

AI Winter — When Expectations Declined

Definition: An AI Winter refers to a period when enthusiasm, investment, or confidence in Artificial Intelligence declined because expected progress did not match practical results.

Contributing difficulties included

- Limited computing resources
- Difficult real-world environments
- Restricted data and knowledge
- Systems that worked only in narrow settings
- Expectations that exceeded actual capabilities

An AI Winter does not mean that AI research completely stopped. It describes a period when confidence in the field declined and progress became more difficult.

From Setbacks to Breakthroughs

AI began making stronger practical progress as several conditions improved together.

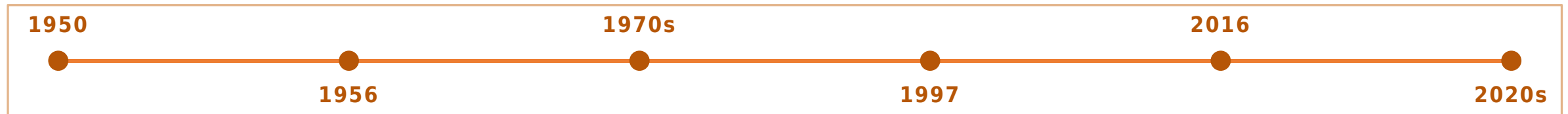
- More powerful computing systems
- Greater availability of data
- Improved algorithms
- Better research methods
- Increasing practical demand for AI solutions

Important milestones: 1997 — Deep Blue 2016 — AlphaGo 2020s —
Modern Generative AI

The evolution of AI is therefore not one single breakthrough, but the result of many improvements working together over time.

AI Timeline — Major Milestones

Year / Period	Milestone
1950	Turing Test
1956	Dartmouth Conference
1970s	AI Winter
1997	Deep Blue defeats Garry Kasparov
2016	AlphaGo defeats Lee Sedol
2020s	Rapid growth of Generative AI



Observe the pattern: Early idea → research field → setbacks → major breakthroughs → modern AI. The timeline shows that AI developed through a long process rather than appearing as a finished technology.

What Changed Across AI Eras?

Earlier AI	Modern AI
Strong reliance on explicitly represented knowledge	Greater use of data-driven approaches
Limited computing resources	Much greater computational capability
Restricted data availability	Large-scale data availability
Mostly specialized research environments	Widespread practical deployment
Narrower system capabilities	Increasingly broad and interactive capabilities

The evolution of AI has been influenced by the combined progress of: Algorithms + Data + Computing + Research + Practical Applications. No single factor explains the transformation of the field.

What Makes an AI System Intelligent?

We have seen what AI is and how it developed. Now we need to examine the capabilities associated with intelligent behavior. Artificial Intelligence may involve:

- 1 Learning**
Acquiring or improving useful patterns
- 3 Planning**
Determining actions toward a goal
- 5 Problem Solving**
Finding suitable ways to achieve an objective

- 2 Reasoning**
Deriving conclusions from available information
- 4 Perception**
Interpreting information from the environment
- 6 Decision Making**
Selecting an appropriate action

These capabilities may work separately or together.

Learning

Definition: Learning is the ability of an AI system to acquire useful patterns or improve its behavior using available information or experience.

Learning can allow a system to

- Identify patterns in information
- Use previous experience
- Improve future predictions
- Adapt its behavior based on new information
- Build useful knowledge for later decisions

Example: A recommendation system can use previous user interactions to improve the content it suggests. The important idea is that the system uses information or experience to improve future behavior.

Reasoning

Definition: Reasoning is the process of using available information to derive conclusions or determine what should follow from a situation.

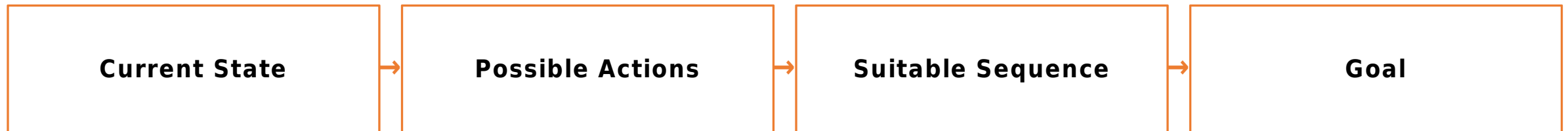


- Connect multiple pieces of information
- Identify relationships
- Draw conclusions
- Evaluate possible interpretations
- Support decisions

Example: A system receives several facts about a situation and combines them to reach a conclusion that is not directly stated in any single input.

Planning

Definition: Planning is the process of determining a sequence of actions that can move a system from its current situation toward a desired goal.



Planning becomes important when the system cannot achieve its objective through a single immediate action.

Example: A navigation system must determine a sequence of movements that takes a user from the current location to the destination. The system is concerned not only with what to do now, but also with what actions should follow.

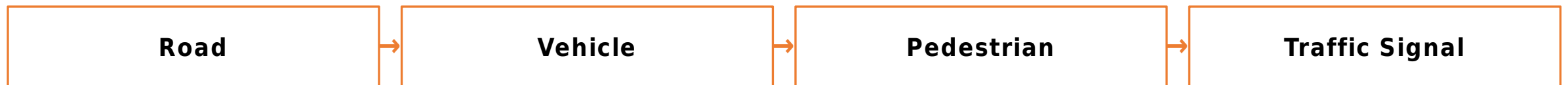
Perception

Definition: Perception is the ability of an AI system to interpret relevant information received from its environment.

The information may come from

- Images
- Audio
- Sensors
- Text
- Other forms of input

The important step is not simply receiving data — the system must determine what that information represents.



Two important AI capabilities are closely related but not identical

Problem Solving and Decision Making

Problem Solving

Finding a suitable way to move from a problem situation toward a desired solution.

Decision Making

Selecting an appropriate action from available alternatives.

Problem solving example

Find possible routes from the current location to the destination

Decision making example

Select the route considered most appropriate under current conditions

The first focuses on finding solutions; the second focuses on choosing an action.

Match each situation with the most suitable capability

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Concept Check — Identify the AI Capability

- 1 Improves future recommendations from previous interactions
- 2 Combines information to reach a conclusion
- 3 Determines a sequence of actions toward a goal
- 4 Interprets information from a camera
- 5 Finds a suitable path toward a destination
- 6 Selects one action from several alternatives

Choices: Learning · Perception · Reasoning · Planning · Problem Solving · Decision Making

Why Classify Artificial Intelligence Systems?

AI systems do not all have the same scope or level of capability. Some are designed for one specific task. Other concepts describe systems with much broader capabilities. This leads to three commonly discussed categories:

- Artificial Narrow Intelligence (ANI)
- Artificial General Intelligence (AGI)
- Artificial Super Intelligence (ASI)

The classification helps us distinguish specialized intelligence from broader forms of intelligence.

Artificial Narrow Intelligence (ANI)

Definition: Artificial Narrow Intelligence (ANI) refers to AI designed to perform a specific task or a limited range of closely related tasks.

- Designed around a defined objective
- Operates within a limited domain
- Can be highly effective at its intended task
- Does not automatically transfer its capability to unrelated tasks
- Represents the practical form of AI widely used today

Examples: Spam filtering · Recommendation systems · Chess engines · Specialized recognition systems

Consider a chess-playing AI

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ANI – Specialized Intelligence

It may

- Analyze possible moves
- Evaluate board positions
- Select strong actions
- Adapt its move selection to the current game

But it cannot automatically

- Diagnose a medical condition
- Teach a programming course
- Manage a factory
- Plan a business strategy

Central idea: A system can be extremely capable within one domain while remaining narrow outside that domain.

Artificial General Intelligence (AGI)

Definition: Artificial General Intelligence (AGI) refers to a hypothetical form of AI with broad capabilities for learning, reasoning, and solving problems across many different domains.

An AGI system is generally expected to

- Learn unfamiliar tasks
- Adapt to new situations
- Apply knowledge across domains
- Transfer useful knowledge between problems
- Handle a broad range of intellectual tasks

The key word is general. The focus is not exceptional performance on one task, but broader intelligent capability across different tasks.

Imagine one system being able to

What Would Make an AI “General”?



The important capability would be the ability to learn, adapt, and apply knowledge across different situations.

Current status: AGI is a hypothetical concept; no confirmed AGI system currently exists.

Artificial Super Intelligence (ASI)

Definition: Artificial Super Intelligence (ASI) refers to a hypothetical form of intelligence that would surpass human intelligence across a broad range of intellectual capabilities.

The concept goes beyond specialized performance or general human-level capability. ASI is imagined as exceeding human abilities in areas such as:

- Reasoning
- Learning
- Problem solving
- Creativity
- Decision making

Because ASI is hypothetical, it also raises questions about control, safety, responsibility, and ethical consequences.

ANI vs AGI vs ASI

Aspect	ANI	AGI	ASI
Scope	Specific / limited	Broad / general	Broad and beyond human capability
Primary idea	Specialized intelligence	General intelligence	Hypothetical superhuman intelligence
Current status	Widely used	Not achieved	Hypothetical
Adaptability	Limited to intended domain	Expected across domains	Expected to exceed humans broadly
Example	Spam filter / chess engine	No confirmed example	No existing example

Key distinction: The classification is based mainly on the scope of capability, not simply on how powerful the system appears.

Classification Scenario – ANI, AGI or ASI?

- A** An AI identifies unwanted messages and separates them from normal email.
- B** An AI can learn unfamiliar tasks, transfer knowledge across domains, and solve a broad range of problems.
- C** An AI is described as exceeding human intellectual performance across virtually all major domains.

A → ANI B → AGI C → ASI

The deciding factor is breadth of capability.

Why Are AI, ML, DL and Data Science Confused?

These four terms frequently appear together, but they represent different ideas. They differ in primary purpose, role of data, methods used, type of problem addressed, and expected output.

Artificial Intelligence	The broader field of intelligent systems
Machine Learning	A major approach for learning from data
Deep Learning	A specialized form of Machine Learning
Data Science	Extracting useful information, patterns, and insights from data

Artificial Intelligence

Definition: Artificial Intelligence (AI) is the broader field concerned with developing systems that perform tasks involving intelligent behavior.

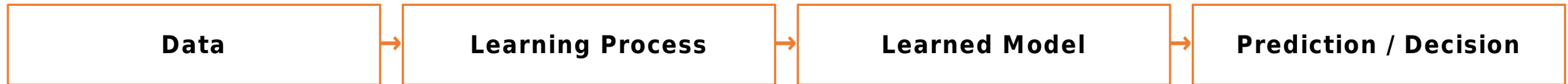
- Learning
- Reasoning
- Planning
- Perception
- Problem solving
- Decision making

AI does not require every system to learn directly from data. It can also use approaches involving rules, logic, search, knowledge representation, and learning.

Example: A game-playing system can use AI techniques to determine actions that move toward a goal.

Machine Learning

Definition: Machine Learning (ML) is an approach in which a system learns useful patterns or relationships from data instead of requiring every behavior to be explicitly programmed.



Machine Learning can support

- Classification
- Prediction
- Pattern recognition
- Recommendation
- Decision support

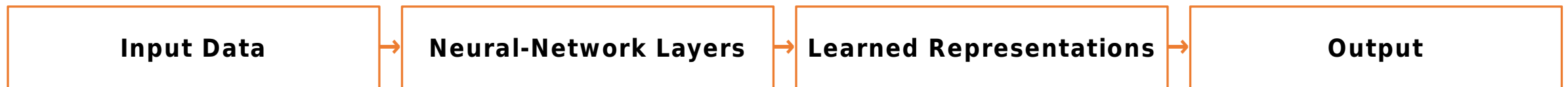
Example: A spam-filtering system can learn patterns from previously identified spam and legitimate messages and use them to classify new messages.

Deep Learning

Definition: Deep Learning (DL) is a specialized area of Machine Learning that uses multi-layered artificial neural networks to learn complex patterns and representations.

Particularly useful for complex information such as

- Images
- Audio
- Video
- Text
- Other high-dimensional data



Example: A face-recognition system can use Deep Learning to learn complex visual patterns from images.

Data Science

Definition: Data Science focuses on extracting useful information, patterns, insights, and knowledge from data.

- Data collection
- Data preparation
- Data exploration
- Statistical analysis
- Visualization
- Predictive methods
- Interpretation and communication

Example: A company may analyze customer and sales data to identify purchasing trends and support business decisions. Data Science can use Machine Learning, but its purpose is broader than building an intelligent model alone.

Consider an online shopping platform

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One Problem — Four Perspectives

Artificial Intelligence

How can the system behave intelligently and recommend useful products?

Machine Learning

How can the system learn customer preferences from previous interactions?

Deep Learning

How can complex patterns in large-scale customer or product information be learned?

Data Science

What useful trends, relationships, and business insights can be extracted from the data?

Important point: One real-world system can use all four areas while each serves a different purpose.

AI vs ML vs DL vs Data Science

Aspect	AI	Machine Learning	Deep Learning	Data Science
Primary focus	Intelligent behavior	Learning from data	Learning complex representations	Extracting insight from data
Typical output	Action / decision	Prediction / classification	Recognition / prediction	Insight / analysis / model
Role of data	May or may not learn directly	Central to learning	Central to training	Central to analysis
Example	Game-playing system	Spam classifier	Face recognition	Sales analysis

These areas overlap, but they should not be treated as interchangeable terms.

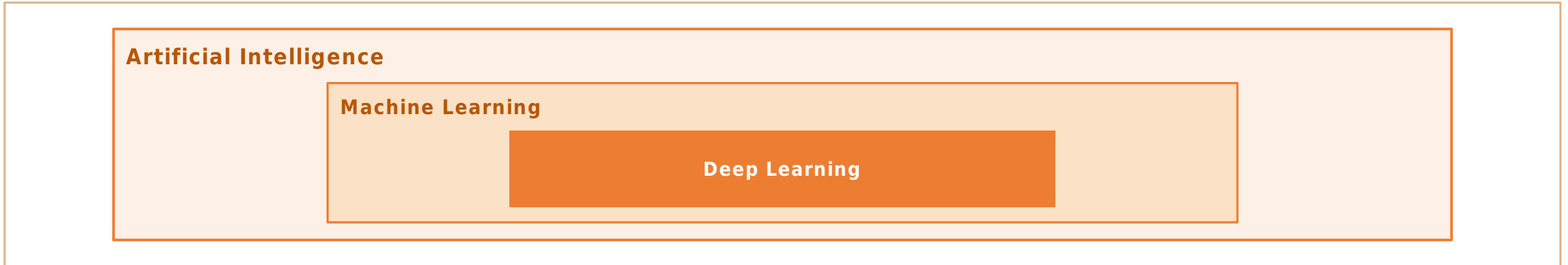
Concept Check — Which Field Fits?

- 1 A company wants to understand why sales changed this quarter.
- 2 A system learns to identify spam messages from historical examples.
- 3 A system identifies objects in an image using a deep neural network.
- 4 A game-playing system determines actions that move toward a goal.

Choices: Deep Learning · Data Science · Machine Learning · Artificial Intelligence

The answer depends on the main objective and method, while recognizing that real systems can combine multiple areas.

How the Four Areas Relate



Data Science overlaps AI + ML through data, analysis & insight

- AI focuses broadly on intelligent behavior
- ML focuses on learning patterns from data
- DL uses deep neural networks to learn complex representations
- Data Science focuses on extracting useful knowledge and insights from data

The boundaries can overlap in real-world systems.

Consider a real-world intelligent application

From Data to Intelligent Action

Step 1 — Data Science	Collect, prepare, explore, and understand the available information
Step 2 — Machine Learning	Learn useful patterns from that information
Step 3 — Deep Learning	Use deep neural networks when complex representations are required
Step 4 — Artificial Intelligence	Use the resulting capabilities within a larger system for intelligent decision or action

Not every AI application requires every stage shown here.

From AI to Intelligent Systems

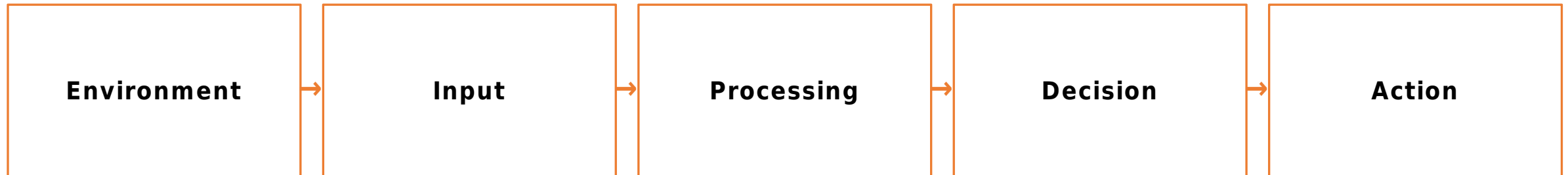
We have now separated the major AI-related fields. The next step is to understand how these ideas become a working system. An intelligent system must do more than contain an AI technique. It must use information to:

- Understand a situation
- Process relevant information
- Determine an appropriate response
- Produce an output or action
- Work toward a defined objective

The focus now moves from AI as a field to AI as a functioning system.

What Is an Intelligent System?

Definition: An intelligent system is a system that uses available information to interpret a situation, make decisions, and produce actions or outputs toward a defined objective.



The system may receive new information after acting and use that information to determine what to do next. This makes an intelligent system interactive and goal-oriented rather than simply a fixed sequence of operations.

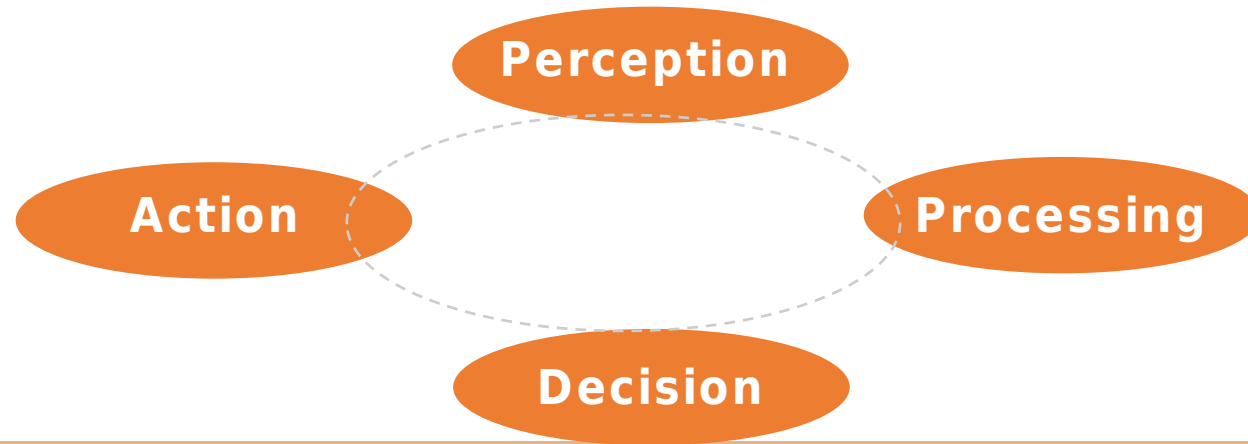
Core Elements of an Intelligent System

Element	Role
Environment	Situation in which the system operates
Input / Sensors	Information available to the system
Processing	Interpretation and analysis of information
Knowledge / Model	Information used to support intelligent behavior
Decision Mechanism	Determines an appropriate response
Action / Output	Produces a result or affects the environment

The exact implementation changes from one application to another. The overall relationship remains: Information → Intelligence → Decision → Action.

Perception → Processing → Decision → Action

A useful way to understand an intelligent system is to follow one complete cycle. The cycle may repeat continuously.



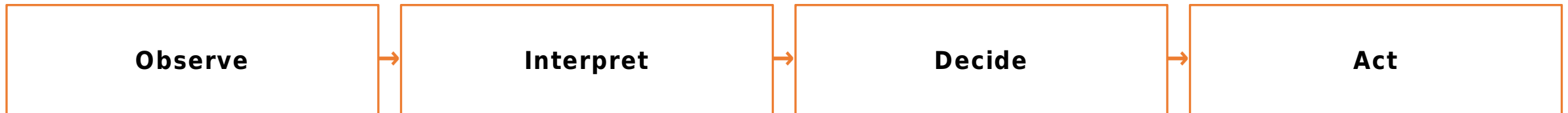
Example: An intelligent road system may detect a change in traffic, interpret the situation, adjust its decision, and then observe the new traffic condition. Intelligence therefore becomes a continuous interaction with the environment.

Imagine a traffic-control system operating at a busy intersection

Scenario — Intelligent Traffic Control

Available information

- Traffic flow
- Vehicle density
- Signal status
- Time-related conditions



The important point is not the particular technology used. It is the relationship between changing information → decision → action → new situation.

Connecting AI Characteristics to System Behavior

The characteristics studied earlier can now be seen inside an intelligent system.

1 Perception
Provides information about the environment

2 Learning
Can improve behavior using information or experience

3 Reasoning
Helps interpret relationships among available information

4 Planning
Identifies possible sequences of actions

5 Problem solving
Works toward a desired result

6 Decision making
Selects an appropriate action

A practical system may use some or all of these capabilities depending on its purpose.

Why Does Artificial Intelligence Need a Lifecycle?

A working AI system is not created in one step. A practical AI solution usually develops through a sequence:



- What problem are we solving?
- What information do we need?
- How should we build the solution?
- Does it work well enough?
- How will it be used?
- Does it continue to perform appropriately?

A strong AI solution therefore depends on more than the model itself.

Stage 1 — Problem Identification

The lifecycle begins by clearly defining what needs to be solved. A useful problem definition identifies:

- The real-world objective
- The intended user or affected system
- The expected outcome
- Available information
- Important constraints

Too broad

“Use AI in agriculture.”

More useful

“Identify possible crop disease from available crop images.”

The second description provides a clear objective that can guide the remaining stages.

Stage 2 — Data and Knowledge Preparation

After defining the problem, the next step is to identify and prepare the information needed to address it. This may involve:

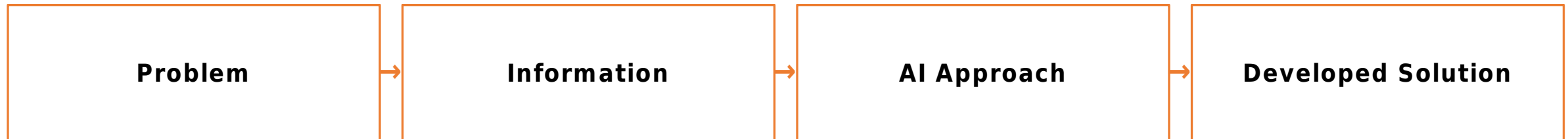
- Collecting relevant information
- Organizing available data
- Removing unsuitable or inconsistent information
- Preparing data in a usable form
- Identifying useful characteristics
- Preparing required knowledge

Example: For crop-disease identification, the system needs relevant crop images that are appropriate for the intended task. The quality and suitability of available information directly affect the usefulness of the resulting AI system.

Stage 3 — AI Approach and Development

The next step is to develop an AI solution that fits the problem. The selected approach depends on:

- Nature of the problem
- Available information
- Required output
- Desired system behavior
- Available resources
- Expected performance



The most complicated method is not automatically the most appropriate one.

Stage 4 — Evaluation and Validation

A developed system must be evaluated before being trusted for practical use. Evaluation asks:

- Does it solve the intended problem?
- How well does it perform?
- Is its behavior reliable?
- What errors does it make?
- What limitations remain?
- Is the performance suitable for the intended use?

Development

Can we build the system?

Evaluation

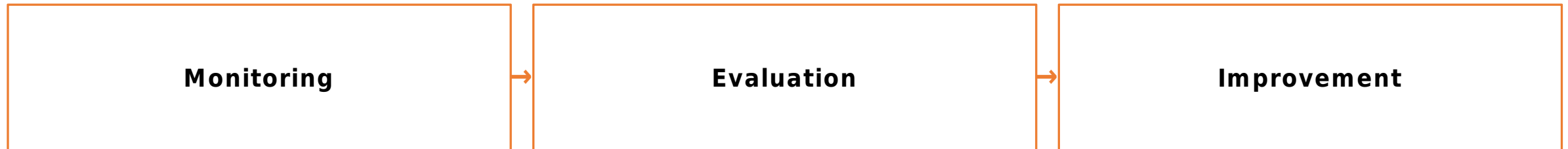
Does the system perform well enough?

A system that runs successfully is not automatically a successful AI solution.

Stage 5 — Deployment and Monitoring

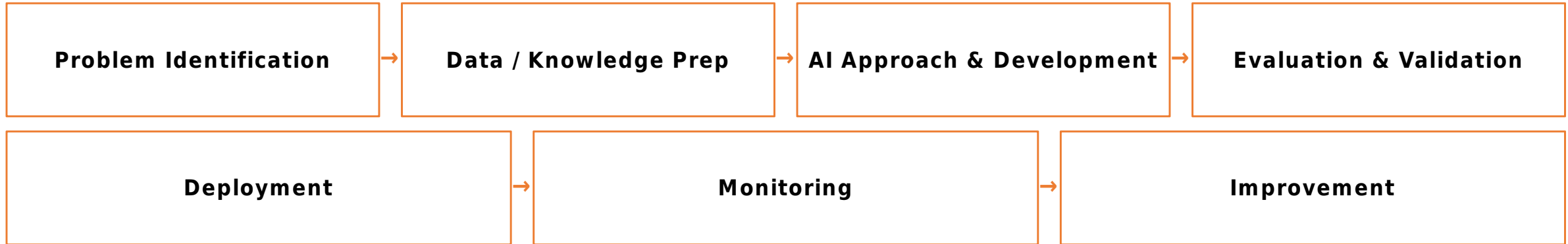
After evaluation, a suitable AI solution can be deployed for its intended use. However, deployment does not necessarily end the lifecycle. Real-world conditions can change:

- New information becomes available
- User behavior changes
- Operating environments change
- System performance changes



A useful AI lifecycle continues after the first deployment.

Complete Artificial Intelligence Lifecycle



Each stage affects the stages that follow. A poorly defined problem can lead to the wrong solution. Poor information can affect system performance. Weak evaluation can result in unsuitable deployment.

The lifecycle provides a structured way to think about AI development from the initial problem to practical operation, looping from Monitoring back into Improvement.

A team has: identified the problem → prepared the information → developed a solution → evaluated its performance

Lifecycle Check — What Happens Next?

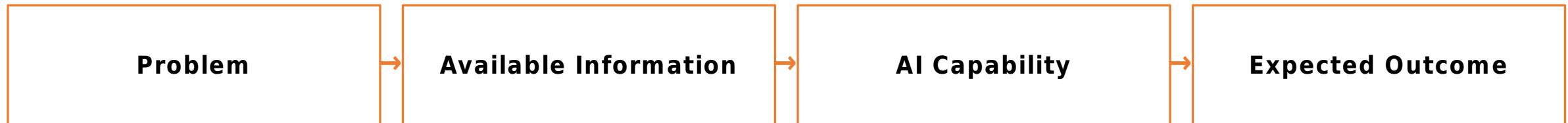
- A** Deploy the validated system for intended use
- B** Remove the original problem definition
- C** Stop evaluating the system permanently
- D** Ignore the evaluation results

The sequence matters because each lifecycle stage prepares the system for the next.

Artificial Intelligence Across Real-World Domains

AI is applied to very different problems across different sectors. The application changes, but common intelligent capabilities can remain relevant. This unit focuses on:

- Healthcare
- Agriculture
- Finance
- Education
- Manufacturing
- Robotics
- Smart Cities



The technology is chosen according to the problem, not the other way around.

Artificial Intelligence in Healthcare and Agriculture

Healthcare

- Medical image analysis
- Pattern identification
- Clinical decision support
- Patient monitoring
- Risk prediction
- Example: analyze medical images to identify patterns requiring further examination

Agriculture

- Crop monitoring
- Disease identification
- Yield prediction
- Resource management
- Agricultural advisory systems
- Example: analyze crop images to identify patterns associated with possible disease

Artificial Intelligence in Finance and Education

Finance

- Fraud detection
- Risk analysis
- Customer behavior analysis
- Financial prediction
- Decision support
- Example: identifying unusual patterns across financial transactions

Education

- Personalized learning
- Intelligent tutoring
- Learning-content recommendations
- Student performance analysis
- Feedback and assistance
- Example: providing learning material based on a student's current interaction and needs

AI in Manufacturing, Robotics and Smart Cities

1

Manufacturing

Quality inspection, Fault detection,
Predictive maintenance, Process monitoring,
Production optimization

2

Robotics

Perception, Navigation, Planning, Decision
making, Physical action

3

Smart Cities

Traffic management, Public transportation,
Energy management, Waste management,
Infrastructure monitoring

Across all three domains, AI connects information, analysis, decisions, and actions to practical problems.

From a Real-World Problem to an AI Problem

We have now seen what AI is, what intelligent systems look like, how an AI lifecycle works, and where AI is applied. But every AI application begins with a problem — usually described informally:

- “Help a vehicle reach its destination.”
- “Detect possible crop disease.”
- “Identify suspicious financial activity.”

The transition: Real-world problem → Formal problem → Computational solution.
This process begins with Artificial Intelligence problem formulation.

What Is AI Problem Formulation?

Definition: AI problem formulation is the process of describing a problem in a structured form so that an AI system can work toward a desired solution.

Initial state	Where the problem begins
Goal state	What successful completion means
Actions / operators	What the system can do
State transitions	How actions change the current state
Constraints	Conditions that limit possible solutions

A well-formulated problem turns a broad objective into something a computational method can work with.

Initial State

“Where are we starting?”

The initial state may describe

- Current location
- Current configuration
- Available resources
- Current system condition
- Information known at the beginning

Example — Route Planning: Initial state: the user's current location. The AI system begins its problem-solving process from this state — without a clearly identified starting state, it is difficult to determine what actions are possible next.

Goal State

“What are we trying to achieve?”

**Route Planning —
Initial state**

Current location

**Route Planning —
Goal state**

Destination

Puzzle — Initial state

Current arrangement of pieces

Puzzle — Goal state

Required final arrangement

The goal does not necessarily describe how to reach the solution — it describes what counts as success. This distinction becomes important when many different paths can lead to the same goal.

Actions and Operators

A problem also needs to describe what the system is allowed to do. An action or operator represents an operation that can change the current state.

Example — Navigation: from one location, possible actions may include

- Take Route A
- Take Route B
- Turn onto another road
- Continue along the current road

Basic relationship: Current State + Action → New State. Without defined actions, the system knows where it is and where it wants to go, but not how it can move toward the goal.

State Transitions

Definition: A state transition describes the change produced when an action is applied to a state.

Current State



Action



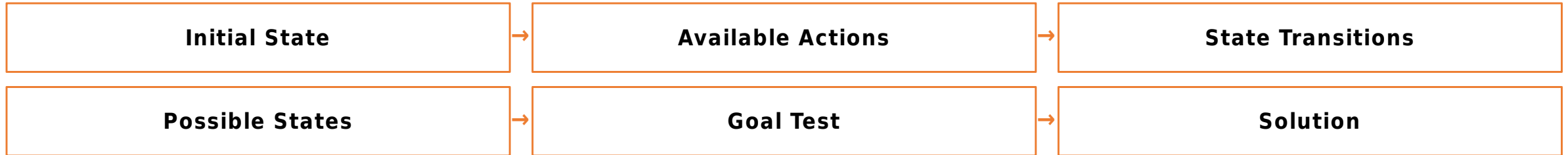
New State

Example

Location A → (Take available road) → Location B

The transition tells the AI system what happens after an action is performed, creating a connection between possible situations. When many states and transitions are connected, they form the structure of the problem's state space.

Complete Problem Formulation



A useful formulation should answer

- Where does the problem start?
- What actions are available?
- How does each action change the situation?
- What state represents success?
- How can we recognize that the goal has been reached?

The quality of the formulation influences everything that comes after it.

Formulate the Problem – Scenario

A delivery vehicle must travel from Warehouse A to Customer D. Available roads connect several locations.

Initial State	Warehouse A
Goal State	Customer D
Actions	Move along an available road
State Transition	Movement changes the vehicle's current location

“Before choosing a solution, what information must the AI system represent?”

This scenario prepares us to represent the possible states and transitions explicitly.

What Is State-Space Representation?

“How can we represent all the possible situations the system may encounter?”

Definition: State-space representation describes a problem using a set of possible states together with the actions and transitions that connect those states.

Initial State



Possible States



Goal State

The state space provides the structure through which a problem-solving method can later search for a suitable solution.

States

Definition: A state is a representation of the relevant situation of the problem at a particular point in time. A state should contain the information needed to understand where the system currently stands.

Navigation

Current location

Puzzle

Current arrangement of pieces

Robot

Current position and relevant environment information

Scheduling

Current assignment of tasks

A state does not need to describe everything in the real world. It should represent the information relevant to solving the problem.

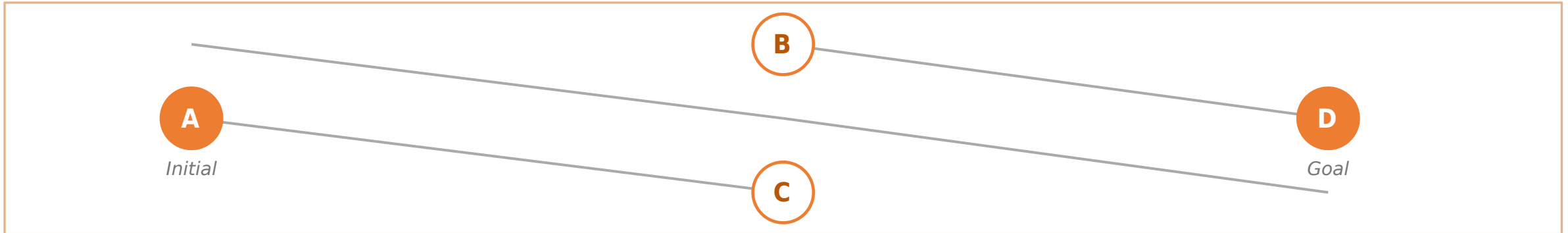
Actions and Transitions in a State Space



- State A is the current situation
- Action 1 changes the situation
- State B represents the result
- Another action produces State C

This relationship creates a network of possible situations. The AI problem-solving process operates on this structure.

State-Space Graph



- Nodes represent states
- Connections represent possible transitions
- Actions determine how transitions occur
- One state is identified as the initial state
- One or more states may satisfy the goal

The graph gives us a structured view of the possible situations the system may encounter.

Initial State → Intermediate States → Goal State

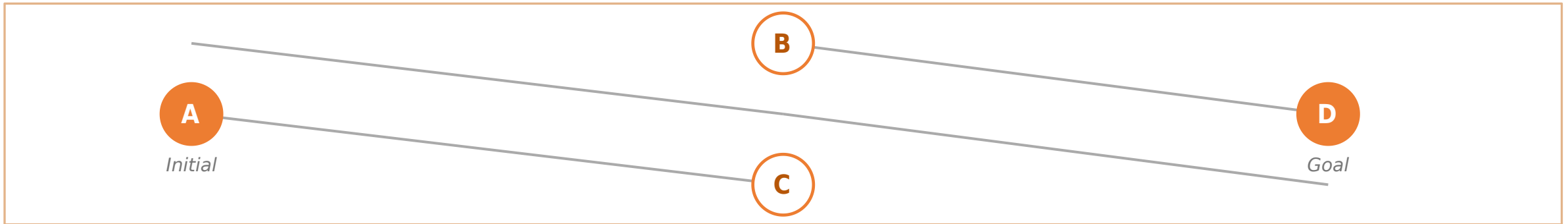


Example

Location A → Location B → Location C → Location D

The system does not necessarily move directly from the initial state to the goal — it may need to pass through several intermediate states. Here, D is the goal, while B and C are intermediate states. This gives us a way to describe the path toward a solution.

State-Space Example — Route Planning



Initial State	A
Goal State	D
Possible routes	$A \rightarrow B \rightarrow D$ or $A \rightarrow C \rightarrow D$

Both represent possible paths through the state space. The AI problem-solving method will later need to determine which path is preferable according to the problem's requirements.

When looking at a state-space diagram, identify the following

Reading a State-Space Representation

- 1. Initial State**
Where does the problem begin?
- 2. Possible States**
What situations can the system reach?
- 3. Actions / Transitions**
How can the system move between states?
- 4. Goal State**
Which state represents successful completion?
- 5. Possible Paths**
What different sequences of transitions can connect the initial state to the goal?

This way of reading a state space will become important when we study actual problem-solving and search techniques later.

Before choosing any solving method, ask

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State-Space Analysis

- Where do we start?
- Where do we want to finish?
- Which states are reachable?
- Which actions create each transition?
- Are multiple paths possible?
- Does every possible path reach the goal?
- Are some states impossible or undesirable?

Key idea: State-space representation makes the structure of the problem visible. It converts an informal problem into a set of states and relationships that a computational method can work with.

Why Do Problem Characteristics Matter?

Not all AI problems behave in the same way. Two problems may both have Initial state → Goal state but still require very different solving approaches. For example:

- One problem may have only a small number of possible states
- Another may have an enormous number
- One may have complete information
- Another may involve uncertainty
- One may have a single clear goal
- Another may involve multiple constraints

Therefore, before selecting a problem-solving approach, we need to understand the characteristics of the problem itself.

Characteristics of an AI Problem

Important characteristics can influence how a problem should be approached.

1 **Size of the problem space**

How many possible states exist

2 **Availability of information**

What the system knows about the problem

3 **Predictability of actions**

Whether an action produces a known result

4 **Goal structure**

Whether the objective is clearly defined

5 **Constraints**

Conditions that limit possible solutions

6 **Complexity**

How difficult it is to identify a useful solution

Problem Structure and Available Information

Situation A

A board game has clearly defined rules, visible states, and known legal actions

Situation B

A system must make a decision using incomplete or changing information

Aspect	Situation A	Situation B
Information	More clearly defined	May be incomplete
Environment	More structured	More variable
Actions	Clearly specified	May depend on conditions
Problem structure	Predictable	Less predictable

The problem characteristics influence how an AI system should approach the solution.

A problem is not defined only by its starting and ending states

Goals, Constraints and Problem Structure

- | | |
|---|---|
| 1 Goal
What counts as success? | 2 Constraints
What actions or states are not allowed? |
| 3 Available information
What does the system know? | 4 Environment
Is the problem static or changing? |
| 5 Possible states
How many situations can be reached? | 6 Solution requirements
Are all valid solutions equally useful? |

These factors determine the structure of the problem and help guide the selection of an appropriate AI-based problem-solving approach.

Consider two problems

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Characteristics and Choice of Approach

Problem A

A structured environment, clearly defined actions, known states, and a precise goal.

Problem B

A changing environment, incomplete information, many possible states, and several constraints.

The same problem-solving approach may not be equally suitable for both.

Understand the problem



Identify its characteristics



Choose an appropriate approach

Compare Two AI Problems

Problem A — Route Planning

- Initial location is known
- Destination is clearly defined
- Possible routes are available
- Different paths may exist
- Travel conditions may affect the preferred solution

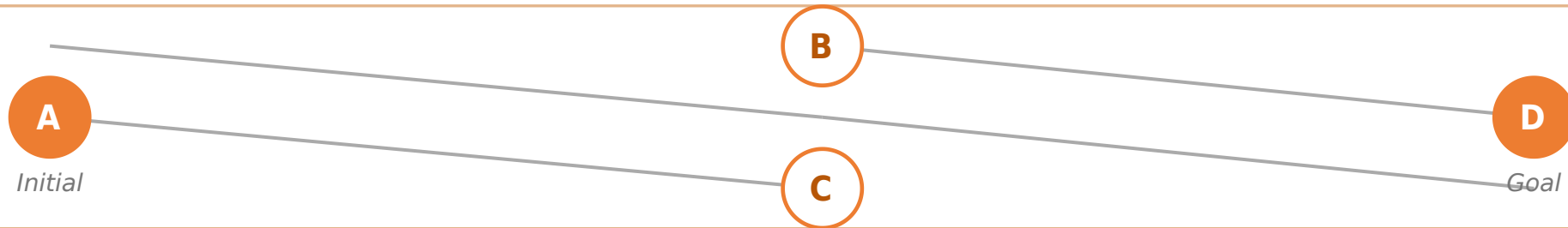
Problem B — Robot Navigation

- Current position must be interpreted
- Environment may change
- Obstacles may appear
- The robot must respond to its surroundings
- The desired action may change as new information arrives

Both problems involve reaching a goal, but their problem characteristics are different. That difference matters when selecting a suitable problem-solving approach.

Search Space

Definition: The search space is the collection of possible states and paths that can be considered while solving a problem.



- States represent possible situations
- Connections represent possible transitions
- Different paths may lead to different states
- The system may need to examine possible paths to reach the goal

A larger search space generally creates a more difficult problem.

Starting from an initial state, several actions may produce different states

Possible States and Candidate Paths

Possible routes

$A \rightarrow B \rightarrow D$ or $A \rightarrow C \rightarrow D$

- Alternative paths
- Repeated states
- Unreachable states
- States that do not lead to the goal

The search space therefore represents the possibilities that may need to be considered, not only the final solution.

Solution Space

Definition: The solution space refers to the set of valid solutions that satisfy the problem's goal and constraints.



Important distinction: Search space describes what may be explored. Solution space describes what satisfies the requirements of the problem. A problem may have many possible states but only a small number of valid solutions.

Search Space vs Solution Space

Aspect	Search Space	Solution Space
Meaning	Possible states and paths that may be considered	Valid solutions satisfying the goal
Scope	Potentially very large	Usually a subset of possibilities
Purpose	Represents what can be explored	Represents what can satisfy the problem
Relationship	Contains candidate possibilities	Exists within the broader possibilities
Example	All possible routes	Routes that reach the destination and satisfy constraints

Think of the relationship as: Search Space → Possible Solutions → Valid Solutions.

Consider a delivery problem: a vehicle can travel through several locations $A \rightarrow B \rightarrow C \rightarrow D$

Search Space or Solution Space?

- Reach the destination
- Exceed a required constraint
- Stop before reaching the destination
- Repeat locations
- Take an undesirable path

1 Which belongs to the search space?

2 Which belongs to the solution space?

1 → All possible routes that can be considered constraints 2 → Routes that satisfy the goal and required

The distinction is simple: possible does not automatically mean valid.

Suppose an AI system finds two valid routes

How Do We Judge an AI Solution?

Route A

Reaches the destination quickly but uses more resources

Route B

Uses fewer resources but takes longer

Both may be valid, so we need a way to evaluate them. Performance measures help answer:

- Is the solution correct?
- Is it efficient?
- Does it satisfy the required constraints?
- How much time or computational effort is required?
- How much resource usage is acceptable?

The best solution depends on what the problem values.

Performance Measures in AI Problem Solving

Performance measures provide criteria for evaluating the quality of a solution or solving process.

1 Correctness
Does the solution achieve the required goal?

2 Efficiency
How much computational effort is required?

3 Time
How quickly can a solution be obtained?

4 Resource use
How much memory or other resources are required?

5 Solution quality
How good is the resulting solution?

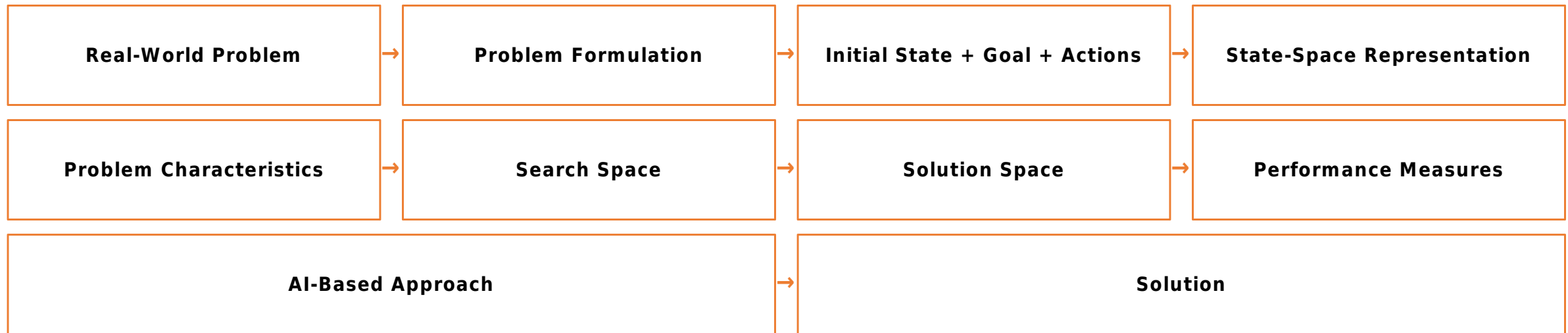
6 Constraint satisfaction
Does the solution respect the problem requirements?

A solution should be evaluated according to the requirements of the problem, not simply whether it exists.

We can now connect the complete Unit I problem-solving process

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From Problem to Solution



A real-world problem must first be understood and represented before an appropriate computational problem-solving approach can be applied. This establishes the foundation for the next unit, where specific AI search techniques can be studied.